

****Fixed parameters****

token: XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX (token is generally defined as a 32-bit verification code)

Token used for testing: 6eb6f069523055a339d71e5b1f6c88cc

If token verification is required, please apply for an oss account and apply by yourself

****Open Platform Overview****

- Growatt's OpenAPI open platform provides standardized RESTful data calling services for authenticated users

- The platform interface is based on the http(s) protocol and uses the OAuth2.0 authentication mechanism.

- Using the API process:

After token verification, please apply for an oss account and apply yourself:

<https://oss.growatt.com/index> (other users worldwide such as Europe),

<https://oss-cn.growatt.com/index> (for Chinese users),

<https://oss-us.growatt.com/index> (North American users)

<https://oss-au.growatt.com/login> (For users in Australia and New Zealand)

****Token usage process:****

1. If a user has been registered on our server monitoring platform, and the device has been successfully monitored, assign the token to the user and obtain data directly, as shown in the following figure:

2. If you have not registered a user on our monitoring platform, then you need to register the server user at the server address corresponding to the token to put the monitoring URL in the server account; then use the API and obtain data as follows:

a. Registered users

b. Create a power station based on registered users

c. Add collectors according to users and power stations

d. Obtain equipment information according to users, power station and collector information, etc.

f. Obtain the detailed data of the device according to the device information (first obtain the device type from the 3.3 interface, and then call different interfaces to obtain the data according to the device type)

You can also specify the token in the account management under the end user settings, as shown below:

****Flow Description****

-TOKEN validity period

Approved: Permanent (tentative)

-Reasons for invalidation of authorization

User cancel authorization

****Address Header:****

Test address:

[http\(s\)://test.growatt.com/v1/](http(s)://test.growatt.com/v1/)

Official address:

Please use the new domain name to visit

(<https://server-cn.growatt.com>-><https://openapi-cn.growatt.com>) (Chinese server)

(<https://server.growatt.com>-><https://openapi.growatt.com>) (foreign server)

(<https://server-us.growatt.com>-><https://openapi-us.growatt.com>) (North American server)

<http://ess-server.atesspower.com> (Era can create)

<http://openapi-au.growatt.com> (for users in Australia and New Zealand)

**** Overall note: ****

- interface supports http, see interface definition

- Using GET/POST, see the interface definition, parameter UTF-8 encoding, URLEncode processing

- Data are encoded using UTF-8

- Support for returning to JSON format

- Return data structure as data: body content, error_code: error code, error_msg: error content

- When calling the interface, you need to add token in the HTTP header: the TOKEN obtained by applying to Growatt

- Time format

Date format: YYYY-MM-DD, example: 2015-04-08

Time format: YYYY-MM-DD HH:mm:ss, example: 2015-04-03 00:01:00

- 示例图：

- 2024

Date	Modifier	Modification content
-----	-----	-----
2024-06-26	Bu Le	Modify max file and return to fault status

- 2024

Date	Modifier	Modification Content
-----	-----	-----
2024-06-21	Bu Le	Add Australian Address

- 2024

Date	Modifier	Modification content
-----	-----	-----
2024-05-07	Bu Le	Modify the relevant address of the vpp interface
http(s)://test.growatt.com/v1/vppRemoteSet Modified to:		
http(s)://test.growatt.com/v1/vppRemoteSetNew		
http(s)://test.growatt.com/v1/vppSet Modified to:		
http(s)://test.growatt.com/v1/vppSetNew		

- 2023

| Date | Modifier | Modification content |

| ----- | ----- | ----- |

| 2023-11-08 | Bu Le | Change the global mix name to SPH|

- 2023

| Date | Modifier | Modification content |

| ----- | ----- | ----- |

| 2023-11-03 | Bu Le | Add two global prompts 10013: The number of each page cannot be greater than 100 |

| 2023-11-03 | Bu Le | 10014: The number of pages cannot be greater than 250 pages |

| Date | Modifier | Modification content |

| ----- | ----- | ----- |

| 2023-10-09 | Bule | All device historical data can only be obtained within 3 months, please store it locally |

- 2023

| Date | Modified By | Modified Content |

| ----- | ----- | ----- |

| 2023-02-17 | Bu Le | Interface 3.4 adds a return parameter plus a netmode (network access method)|

- 2022

| Date | Modified By | Modified Content |

| ----- | ----- | ----- |

| 2022-09-22 | Bu Le | Document description modification Users can apply for token by themselves|

- 2022

| Date | Modified By | Modified Content |

| ----- | ----- | ----- |

| 2022-08-30 | Bu Le | To change the token application process, you can apply by yourself through the oss account|

- 2022

| Date | Modified By | Modified Content |

| ----- | ----- | ----- |

| 2022-07-11 | Bu Le | Add a setting item to the MIN setting interface: pre_pto 0: Off, 1: On|

- 2022

| Date | Modified By | Modified Content |

| ----- | ----- | ----- |

| 2022-06-29 | Bu Le | Inverter alarm added time limit date: a certain date|

| 2022-06-29 | Bu Le | max alarm return time limit date: a certain date|

- 2022

| Date | Modified By | Modified Content |

| ----- | ----- | ----- |

| 2022-06-07 | Bu Le | 3.5 Added return parameters dtc: device code, model: device
model,normalPower: rated power|

- 2022

| Date | Modified By | Modified Content |

| ----- | ----- | ----- |

| 2022-05-011 | Bu Le | Delete the return value of all equipment basic information, current
power Power, eToday, eTotal|

| 2022-05-011 | Bu Le | Add interface: obtain the return value of the real-time data of the
environmental detector according to the collector SN and the address of the
environmental detector, etodayRadiation, daily radiation|

- 2022

| Date | Modified By | Modified Content |

| ----- | ----- | ----- |

| 2022-05-07 | Bu Le | Delete the return value of max basic information, the current power
Power |

- 2022

| Date | Modified By | Modified Content |

| ----- | ----- | ----- |

| 2022-04-11 | Bu Le | Added Min, return parameters for latest data and historical data |
| 2022-04-11 | Bu Le | Added Min, page parameter interface |
- 2019

| Date | Modifier | Modify Content|
| -----|
| 2019-03-05 | Zhu Xiaohai | Added Mix, Spa Device Data and Basic Information Interface|
| 2019-06-12 | Zhu Xiaohai | New parameters for the push interface (Max, Min) |
| 2019-12-05 | Zhu Xiaohai | Added Min (Tlx) data and basic information interface |

- 2018

Date	Modifier	Content
2018-01-10	Zhu Xiaohai	Registered User Interface with New National Selection Value
and New Time Zone Parameters		
2018-03-27	Zhu Xiaohai	Add a new interface to obtain the information of the power
station to which the equipment serial number belongs according to the equipment serial		
number		
2018-04-03	Zhu Xiaohai	New Access to Basic Information Interface of Energy Storage
Machine Based on Equipment Sequence Number		
2018-08-26	Zhu Xiaohai	Data and Basic Information Access Interface for New Max
Devices

- 2017

Date	Modifier	Content
2017-01-26	Zhu Xiaohai	Modify the parameter device_id to device_sn
2017-03-28	Zhu Xiaohai	Power Station List Interface Return Parameters Add latitude_d,
latitude_f, Power Station Modification Interface Request Parameters Add latitude_d,		
latitude_f		
2017-04-10	Zhu Xiaohai	Added List Interface and Detailed Data Interface for Accessing
Smart Meter and Environmental Detector Equipment |

| 2017-12-05 | Zhu Xiaohai | Added information interface and data information needed to acquire data that is pushed to customers actively |

- 2016

| Date | Modifier | Content |

| ----- | ----- | ----- |

| 2016-08-26 | Zhu Xiaohai | Document Publication |

| 2016-08-30 | Zhu Xiaohai | New user name and collector verification interface examples, improve the document structure |

| 2016-11-21 | Zhu Xiaohai | The only token in the HTTP request header needs to contact Gurivat New Energy Co., Ltd. to obtain access Token interface to stop using |

| 2016-11-22 | Zhu Xiaohai | New user_plant_list interface based on user ID pagination |

| 2016-11-30 | Zhu Xiaohai | Remove API interfaces that are no longer in use

| 2016-12-20 | Zhu Xiaohai | Detailed data acquisition interface, alarm data acquisition interface and parameter setting interface for new energy storage machine |

****remarks****

| Error Code | Error Interpretation |

| ----- | ----- | ----- |

| 0 | Normal (General) |

| 10011 | No privilege access (generic)|

| 10012 | Access Frequency Limitation of 5 Minutes/Time (Universal)|

| 10013 | The number per page cannot be greater than 100 (general) |

| 10014 | The number of pages cannot be greater than 250 pages (general) |

| -1 | Please use the new domain name to access (https://server-cn.growatt.com->https://openapi-cn.growatt.com) (https://server.growatt.com->https://openapi.growatt.com)(https://server-us.growatt.com->https://openapi-us.growatt.com) |

****A brief description:****

- User registered interface

****Request URL: ****

- ` http(s)://test.growatt.com/v1/user/user\_register `

****Request method: ****

- post

****Parameters: ****

Parameter Name	Whether Required	Parameter Value	Description
----	----	----	----
user_name	Yes	string	username
user_password	Yes	string	Password
user_email	Yes	string	Register Email
user_type	Yes	int	User Type (1 for end customers)
user_country	Yes	string	User Country [country list](https://www.showdoc.com.cn/dfsdfd/9187729846853701 "country list")
agentCode	No	string	Installer code
user_tel	No	int	User Phone
time	No	string	User Time Zone

****Back to example****

```

```
{
 "data": {
 "c_user_id": 54
 },
 "error_code": 0,
```

```
"error_msg": ""
}
` ``
```

**\*\*Return parameter description\*\***

| Parameter Name | Type   | Description                                                                                                                                                                                                                 |
|----------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ----           | ----   | ----                                                                                                                                                                                                                        |
| c_user_id      | int    | End User ID                                                                                                                                                                                                                 |
| error_code     | int    | 0: Normal return, 10001: System error, 10002: Username or password is empty, 10003: Username already exists, 10004: User mailbox is empty, 10006: User type is empty, 10008: Token is empty, 10014 : Installer coding error |
| error_msg      | string | Error Message Prompt                                                                                                                                                                                                        |

**\*\*Remarks\*\***

**\*\*A brief description:\*\***

- Verify that the username is duplicated

**\*\*Request URL: \*\***

- ` http(s)://test.growatt.com/v1/user/check\_user `

**\*\*Request method: \*\***

- post

**\*\*Parameters: \*\***

| Parameter Name | Whether Required | Type   | Description |
|----------------|------------------|--------|-------------|
| -----          | :---             | :----- | -----       |
| user_name      | Yes              | string | username    |

**\*\*Back to example\*\***

```
```\n{\n  "data": "",\n  "error_code": 0,\n  "error_msg": ""\n}\n```\n
```

****Return parameter description****

| Parameter Name | Type | Description |
|----------------|--------|---|
| :----- | :----- | :----- |
| error_code | int | 0: Normal return, 10001: Server exception, 10002: Username is empty, 10003: Username already exists |
| error_msg | string | Error message prompt |

****Remarks****

- Get the frequency once every 5 minutes

****A brief description:****

- Get the interface of the end user list under the merchant

****Request URL: ****

- ` http(s)://test.growatt.com/v1/user/c\_user\_list `

****Request method: ****

- get

****Parameters: ****

| Parameter Name | Whether Required | Type | Description |
|----------------|------------------|------|-------------|
|----------------|------------------|------|-------------|

| :---- | :--- | :----- | :----- |
|-------|------|--------|--------|
|-------|------|--------|--------|

| | | | |
|------|----|-----|-----------------------|
| page | No | int | page number,default 1 |
|------|----|-----|-----------------------|

| | | | |
|---------|----|-----|---|
| perpage | No | int | Number of pages per page, default 20, max 100 |
|---------|----|-----|---|

****Back to example****

```

```
{
 "data": {
 "count": 2,
 "c_user": [
 {
 "c_user_id": 1,
 "c_user_name": "admin",
 "c_user_email": "imluobao@163.com",
 "c_user_tel": "",
 "c_user_regtime": "2018-02-04 09:46:50"
 }
]
 },
 "error_code": 0,
 "error_msg": ""
}
```

```

****Return parameter description****

| Parameter Name | Type | Description |
|----------------|------|-------------|
|----------------|------|-------------|

| : | ---- | : | ----- | : | ----- | : |
|---|------|---|-------|---|-------|---|
|---|------|---|-------|---|-------|---|

| | | |
|-----------|-----|-------------|
| c_user_id | int | End User ID |
|-----------|-----|-------------|

| | | |
|-------------|--------|---------------|
| c_user_name | string | End User Name |
|-------------|--------|---------------|

| | | |
|--------------|--------|-----------------------------|
| c_user_email | string | End User Registration Email |
|--------------|--------|-----------------------------|

| | | |
|----------------|--------|----------------------------|
| c_user_regtime | string | end user registration date |
|----------------|--------|----------------------------|

| | | |
|------------|-----|----------------|
| c_user_tel | int | End User Phone |
|------------|-----|----------------|

| | | |
|-------|-----|---------------------------|
| count | int | Total number of end users |
|-------|-----|---------------------------|

| | | |
|------------|-----|---------------------------------------|
| error_code | int | 0: Normal return, 10001: System error |
|------------|-----|---------------------------------------|

| | | |
|-----------|--------|----------------------|
| error_msg | string | Error message prompt |
|-----------|--------|----------------------|

****Remarks****

- Get the frequency once every 5 minutes
- This interface is only allowed to call 10 times a day

****A brief description:****

- Add the interface of the power station

****Request URL: ****

- ` http(s)://test.growatt.com/v1/plant/add `

****Request method: ****

- post

****Parameters: ****

| Parameter name | Is it required | Type | Description |
|----------------|----------------|------|-------------|
|----------------|----------------|------|-------------|

| : | ---- | : | --- | : | ----- | : | ----- | : |
|---|------|---|-----|---|-------|---|-------|---|
|---|------|---|-----|---|-------|---|-------|---|

|c_user_id |Yes |int |User ID to which the power station belongs |

|name |Yes |string |power station name |

|peak_power |Yes |float |peak power (kWp) |

|latitude_f |No |string | Area [country
list](<https://www.showdoc.com.cn/dfsdf/9187729846853701> "country list") |

|latitude_d |No |string |Installer code |

|currency |No |boolean |currency unit|

|longitude |No |boolean |longitude |

|latitude |No |boolean |latitude |

|timezone_id |No |int |The time zone code of the data display |

|plant_type |No |int |Power station type: 0: household power station, 1: industrial and
commercial power station, 2: poverty alleviation power station |

|createDate |No |string |Created time |

|formulaMoney |No |float |Electricity price |

|city |No |string |city |

|plantAddress |No |string |City |

****Call example****

| Parameter name | Is it required | Value | Description |
|----------------|----------------|----------|---|
| :---- | :--- | :----- | :----- |
| c_user_id | Yes | 74 | User ID to which the power station belongs |
| name | is | bole66 | power station name |
| peak_power | Yes | 20 | Peak power (kWp) |
| latitude_f | No | Thailand | Region [Country
List](https://www.showdoc.com.cn/v1/1313587669914582 "Country List") |
| latitude_d | no | AFLF6 | Installer code |
| currency | No | \$ | Currency unit |
| longitude | no | 30 | longitude |

|latitude |no |22 |latitude |

|timezone_id |No |8 |Time zone code for data display |

|plant_type |No |1 |Power station type: 0: Household power station, 1: Industrial and commercial power station, 2: Poverty alleviation power station |

|createDate |No |2022-10-13 |Creation time |

|formulaMoney |No |1.4 |Electricity Price |

|city |no |bankok |city |

|plantAddress |No |bankok |City |

...

{

"c_user_id": "74",

"name": "bole66",

"peak_power": "20",

"latitude_f": "Thailand",

"latitude_d": "AFLF6",

"currency": "\$",

"longitude": "30",

"latitude": "22",

"timezone_id": "8",

"plant_type": "1",

"createDate": "2022-10-13",

"formulaMoney": "1",

"city": "bankok",

"plantAddress": "bankok",

}

...

****Return to Example****

...

{


```

    "data": {
        "plant_id": 24832
    },
    "error_code": 0,
    "error_msg": ""
}
...

```

****Return parameter description****

| Parameter name | Type | Description |
|----------------|--------|----------------------|
| plant_id | int | Plant ID |
| error_code | int | error code |
| error_msg | string | Error message prompt |

```

plant_id	error_code	error_msg
77	0	Return normally

```

```

...

{
    "error_msg": "",
    "data": {
        "plant_id": 77
    },
    "error_code": 0
}
...

```

|error_code|error_msg|

|:----- |:-----|----- |

|10001|System error|

\\ \

{

"error_msg": "error_system error",

"data": "",

"error_code": 10001

}

\\ \

|error_code|error_msg|

|:----- |:-----|----- |

|10002|Username ID is empty|

\\ \

{

"error_msg": "error_User ID is empty",

"data": "",

"error_code": 10002

}

\\ \

|error_code|error_msg|

|:----- |:-----|----- |

|10003|The power station name is empty|

\\ \

{

"error_msg": "error_Plant name is empty",

"data": "",

"error_code": 10003

}

...

|error_code|error_msg|

|:----- |:-----|----- |

|10004|Peak power is empty|

...

{

"error_msg": "error_Peak power is empty",

"data": "",

"error_code": 10004

}

...

|error_code|error_msg|

|:----- |:-----|----- |

|10005|User does not exist|

...

{

"error_msg": "error_User does not exist",

"data": "",

"error_code": 10005

}

...

|error_code|error_msg|

|:----- |:-----|----- |

|10006|The power station name already exists under this user|

...

{

```
"error_msg": "error_Plant name already exists under this user",  
"data": "",  
"error_code": 10006  
}  
` ``
```

****Remark****

- The acquisition frequency is once every 5 minutes

****A brief description:****

- Modify the interface of the power station

****Request URL: ****

- ` http(s)://test.growatt.com/v1/plant/modify `

****Request method: ****

- post

****Parameters: ****

| Parameter Name | Whether Required | Type | Description |
|----------------|------------------|------|-------------|
|----------------|------------------|------|-------------|

| :--- | :--- | :----- | :----- |
|------|------|--------|--------|
|------|------|--------|--------|

| | | | |
|-----------|-----|-----|------------------------------------|
| c_user_id | Yes | int | User id to which the plant belongs |
|-----------|-----|-----|------------------------------------|

| | | | |
|----------|-----|-----|------------------|
| plant_id | Yes | int | Power Station ID |
|----------|-----|-----|------------------|

| | | | |
|------|-----|--------|--------------------|
| name | Yes | string | power station name |
|------|-----|--------|--------------------|

| | | | |
|------------|----|--------|--|
| latitude_f | No | string | Area [country list](https://www.showdoc.com.cn/dfsdf/9187729846853701 "country list") |
|------------|----|--------|--|

| | | | |
|------------|----|--------|----------------|
| latitude_d | No | string | Installer Code |
|------------|----|--------|----------------|

| | | | |
|------------|----|---------|------------------|
| peak_power | No | boolean | peak power (kWp) |
|------------|----|---------|------------------|

| | | | |
|----------|----|---------|---------------|
| currency | No | boolean | currency unit |
|----------|----|---------|---------------|

|longitude |No |boolean |Longitude |

|latitude | No |boolean | latitude |

|timezone_id |No |int |Time zone code for data display: East time zone: 1~12 West time zone: -1~-12 |

|plant_type |No |int |Power station type: 0: household power station, 1: industrial and commercial power station, 2: poverty alleviation power station |

****Call example****

|Parameter name|Is it required|Value|Description|

|:---- |:---|:----- |----- |

|c_user_id |Yes |74 |User ID to which the power station belongs |

|plant_id |Yes |77 |Plant ID |

|name |Yes |20 |Power station name |

|latitude_f | No |Thailand | Region [Country List](<https://www.showdoc.com.cn/v1/1313587669914582> "Country List")|

|latitude_d |no |AFLF6 |Installer code |

|peak_power |No |20 |Peak power (kWp) |

|currency |No |\$ |Currency unit|

|longitude |no |30 |longitude |

|latitude |no |23 |latitude |

|timezone_id |No |8 |Time zone code displayed by data: East time zone: 1~12 West time zone: -1~-12 |

|plant_type |No |0 |Power station type: 0: Household power station, 1: Industrial and commercial power station, 2: Poverty alleviation power station |

````

{

"c\_user\_id": "74",

"plant\_id": "77",

"name": "bole66",

"latitude\_f": "Thailand",

"latitude\_d": "AFLF6",

"peak\_power": "20",

```
"currency": "$",
"longitude": "30",
"latitude": "23",
"timezone_id": "8",
"plant_type": "0"
}
...
```

**\*\*Return to Example\*\***

```
...
{
 "data": "",
 "error_code": 0,
 "error_msg": ""
}
...
```

**\*\*Return parameter description\*\***

Parameter name	Type	Description
error_code	int	error code
error_msg	string	Error message prompt

error_code	error_msg
0	Return normally

```
...
{
 "data": "",

```

```
"error_code": 0,
"error_msg": ""
}
...
```

```
|error_code|error_msg|
|:----- |:-----|----- |
|10001|System error|
```

```
...

{
"error_msg": "error_System error",
"data": "",
"error_code": 10001
}
...
```

```
|error_code|error_msg|
|:----- |:-----|----- |
|10002|Username ID is empty|
```

```
...

{
"error_msg": "error_User ID is empty",
"data": "",
"error_code": 10002
}
...
```

```
|error_code|error_msg|
|:----- |:-----|----- |
|10003|The power station name is empty|
```

...

```
{
"error_msg": "error_Plant name is empty",
"data": "",
"error_code": 10003
}
```

...

```
|error_code|error_msg|
|:----- |:-----|----- |
|10004|Peak power is empty|
```

...

```
{
"error_msg": "error_Peak power is empty",
"data": "",
"error_code": 10004
}
```

...

```
|error_code|error_msg|
|:----- |:-----|----- |
|10005|User does not exist|
```

...

```
{
"error_msg": "error_User does not exist",
"data": "",
"error_code": 10005
}
```

...

```
|error_code|error_msg|
```



|:----- |:-----|----- |

|10006|The power station name already exists under this user|

`, `

{

"error\_msg": "error\_Plant name already",

"data": "",

"error\_code": 10006

}

`, `

|error\_code|error\_msg|

|:----- |:-----|----- |

|10007|The power station does not exist|

`, `

{

"error\_msg": "error\_Plant does not exist",

"data": "",

"error\_code": 10007

}

`, `

**\*\*Remark\*\***

- This interface is only allowed to be called 10 times a day

**\*\*A brief description:\*\***

- Delete the interface of the power station

**\*\*Request URL: \*\***

- ` http(s)://test.growatt.com/v1/plant/delete `

**\*\*Call example\*\***

Parameter name	Is it required	Value	Description
----------------	----------------	-------	-------------

:	----	:	---	:	-----	:	-----	:
---	------	---	-----	---	-------	---	-------	---

plant_id	Yes	77	Plant ID					
----------	-----	----	----------	--	--	--	--	--

...

{

"plant\_id": "77",

}

...

**\*\*Return to Example\*\***

...

{

"data": "",

"error\_code": 0,

"error\_msg": ""

}

...

**\*\*Return parameter description\*\***

Parameter name	Type	Description
----------------	------	-------------

:	-----	:	-----	:	-----	:
---	-------	---	-------	---	-------	---

error_code	int	error code				
------------	-----	------------	--	--	--	--

error_msg	string	Error message prompt				
-----------	--------	----------------------	--	--	--	--

error_code	error_msg					
------------	-----------	--	--	--	--	--

|:----- |:-----|----- |

|0|Return normally|

```

{

 "data": "",

 "error_code": 0,

 "error_msg": ""

}

```

|error\_code|error\_msg|

|:----- |:-----|----- |

|10001|System error|

```

{

 "error_msg": "error_system error",

 "data": "",

 "error_code": 10001

}

```

|error\_code|error\_msg|

|:----- |:-----|----- |

|10002|The power plant ID is empty|

```

{

 "error_msg": "error_plant ID is empty",

 "data": "",

"error_code": 10002

}

...

|error_code|error_msg|

|:-----|-----|

|10004|The power station does not exist|

...

{

"error_msg": "error_plant does not exist",

"data": "",

"error_code": 10004

}

...

****Remark****

****A brief description:****

- Get the interface of a user's power station list

****Request URL: ****

- ` http(s)://test.growatt.com/v1/plant/user\_plant\_list `

****Request method: ****

- post

****Parameters: ****

|Parameter Name|Whether Required|Type|Description|

|:----|:---|:-----|-----|

|user_name|Yes|string|User Account|

|page|No|int|page number,default 1|

|perpage|No|int|Number of pages per page, default 20, max 100|

****Call example****

|Parameter name|Is it required|Value|Description|

|:----|:---|:-----|-----|

|user_name|is|1236985|user account|

|page|No|1|Page number, default 1|

|perpage|No|20|Number per page, default 20, maximum 100|

...

{

"user_name": "1236985",

"page": "1",

"perpage": "20"

..

}

...

****Return to Example****

...

{

"data": {

"count": 1,

"plants": [

{

"status": 4,

"locale": "en-US",

"total_energy": 15.699999809265137,

"operator": "API interface testing manufacturer",

"country": "China",

```

        "city": "Shenzhen",
        "current_power": "",
        "create_date": "2018-12-12",
        "image_url": "2.png",
        "plant_id": 24765,
        "name": "API interface test power station",
        "installer": "API interface testing manufacturer",
        "user_id": 33,
        "longitude": "113.9",
        "latitude": "22.6",
        "peak_power": 20,
        "latitude_d": null,
        "latitude_f": null
    },
]
},
"error_code": 0,
"error_msg": ""
}
...

```

****Return parameter description****

| Parameter name | Type | Description |
|----------------|--------|--|
| plant_id | int | Plant ID |
| name | string | Power station name |
| user_id | int | User ID to which the power station belongs |
| country | string | country name |
| longitude | string | longitude |
| latitude | string | latitude |

|peak_power |string |Peak power (kWp)
|create_date |string |Website creation date
|installer |string |Installer name
|current_power |string |Current power (kW)
|total_energy |string |Cumulative power generation (kWh)
|count |int |Total number of power stations
|error_code |int |error code
|error_msg |string |Error message prompt

|error_code|error_msg|

|:----- |:-----|----- |

|0|Return normally|

````

```
{
 "data": {
 "count": 1,
 "plants": [
 {
 "status": 4,
 "locale": "en-US",
 "total_energy": 15.699999809265137,
 "operator": "API Interface Test Vendor",
 "country": "China",
 "city": "Shenzhen",
 "current_power": "",
 "create_date": "2018-12-12",
 "image_url": "2.png",
 "plant_id": 24765,
 "name": "API interface test station",
 "installer": "API Interface Test Vendor",
 "user_id": 33,
```

```

 "longitude": "113.9",
 "latitude": "22.6",
 "peak_power": 20,
 "latitude_d": null,
 "latitude_f": null
 },
]
},
"error_code": 0,
"error_msg": ""
}
...

|error_code|error_msg|
|:----- |:-----|----- |
|10001|System error|
...

{
 "data": "",
 "error_code": 10001,
 "error_msg": "error_system error"
}
...

Remark

```

- The acquisition frequency is once every 5 minutes
- This interface is only allowed to be called 10 times a day

**\*\*A brief description:\*\***

- Get the interface to all user power station lists

**\*\*Request URL: \*\***



- ` http(s)://test.growatt.com/v1/plant/list `

**\*\*Request method: \*\***

- get

**\*\*Parameters: \*\***

Parameter Name	Whether Required	Type	Description
----------------	------------------	------	-------------

:----	:---	:-----	-----
-------	------	--------	-------

page	No	int	page number,default 1
------	----	-----	-----------------------

perpage	No	int	Number of pages per page, default 20, max 100
---------	----	-----	-----------------------------------------------

search_type	No	string	search type
-------------	----	--------	-------------

search_keyword	No	string	search keywords
----------------	----	--------	-----------------

**\*\*Call example\*\***

Parameter name	Is it required	Value	Description
----------------	----------------	-------	-------------

:----	:---	:-----	-----
-------	------	--------	-------

page	No	1	Page number, default 1
------	----	---	------------------------

perpage	No	100	Number per page, default 20, maximum 100
---------	----	-----	------------------------------------------

search_type	No		Search type
-------------	----	--	-------------

search_keyword	No	test	Search keyword
----------------	----	------	----------------

```\n

{\n

"page": "1",\n

"perpage": "100",\n

"search_type": "",\n

"search_keyword": "test"\n

}\n

```\n

**\*\*Return to Example\*\***

...

```
{
 "data": {
 "count": 1,
 "plants": [
 {
 "status": 1,
 "locale": "en-US",
 "total_energy": 0,
 "operator": "0",
 "country": "China",
 "city": "0",
 "current_power": "",
 "create_date": "2016-09-15",
 "image_url": null,
 "plant_id": 1185,
 "name": "9",
 "installer": "0",
 "user_id": 31,
 "longitude": "",
 "latitude": "",
 "peak_power": 0,
 "latitude_d": null,
 "latitude_f": null
 }
]
 },
 "error_code": 0,
 "error_msg": ""
}
```

...

**\*\*Return parameter description\*\***

|Parameter name|Type|Description|

|:----- |:-----|----- |

|plant\_id |int |Plant ID

|name |string |Power station name

|user\_id |int |User ID to which the power station belongs

|country |string |country name

|longitude |string |longitude

|latitude |string |latitude

|peak\_power |string |Peak power (kWp)

|create\_date |string |Website creation date

|installer |string |Installer name

|current\_power |string |Current power (W)

|total\_energy |string |Cumulative power generation (kWh)

|count |int |Total number of power stations

|latitude\_d |string |Agent name

|latitude\_f |string |Agency area

|error\_code |int |error code

|error\_msg |string |Error message prompt

|error\_code|error\_msg|

|:----- |:-----|----- |

|0|Return normally|

...

{

  "data": {

    "count": 1,

    "plants": [

```

 {
 "status": 1,
 "locale": "en-US",
 "total_energy": 0,
 "operator": "0",
 "country": "China",
 "city": "0",
 "current_power": "",
 "create_date": "2016-09-15",
 "image_url": null,
 "plant_id": 1185,
 "name": "9",
 "installer": "0",
 "user_id": 31,
 "longitude": "",
 "latitude": "",
 "peak_power": 0,
 "latitude_d": null,
 "latitude_f": null
 }
]
},
"error_code": 0,
"error_msg": ""
}
...

|error_code|error_msg|
|:----- |:-----|----- |
|10001|System error|
...

{

```

```
"data": "",
"error_code": 10001,
"error_msg": "error_system error"
}
` ` `
```

#### **\*\*Remark\*\***

- The acquisition frequency is once every 5 minutes
- This interface is only allowed to be called 10 times a day

#### **\*\*A brief description:\*\***

- Interface to obtain basic information about a power station

#### **\*\*Request URL: \*\***

- ` http(s)://test.growatt.com/v1/plant/details `

#### **\*\*Request method: \*\***

- get

#### **\*\*Parameters: \*\***

| Parameter Name            | Whether Required | Type | Description      |
|---------------------------|------------------|------|------------------|
| :----  :--- :-----  ----- |                  |      |                  |
| plant_id                  | Yes              | int  | Power Station ID |

#### **\*\*Back to example\*\***

` ` `

```
{
 "data": {
 "address1": "",
 "jurisdictionorganization": "",
 "address2": "",
 "installed_dc_capacity": "",
 "city": "Shenzhen",
 "timezone": "GMT+8",
 "designerorganization": "",
 "installercontact": "",
 "create_date": "2018-12-12",
 "image_url": "2.png",
 "description": "",
 "longitude": "113.9",
 "fixed_azimuth": "",
 "oftakercontact": "",
 "status": "",
 "postal": "",
 "weathersensor_num": "",
 "weathersensor_man": "",
 "operatorcontact": "",
 "country": "China",
 "plant_type": "",
 "fixed_tilt": "",
 "inverters": [
 {
 "inverter_md": "",
 "inverter_num": 1,
 "inverter_man": "Growatt"
 }
],
 },
}
```

```
"installerorganization": "",
"jurisdictioncontact": "",
"installed_ac_capacity": "",
"latitude": "22.6",
"financiercontact": "",
"ownerorganization": "rmb",
"locale": "en_US",
"designercontact": "",
"state": "",
"weather_type": "",
"currency": "rmb",
"installed_panel_area": "",
"name": "API interface test power station",
"user_id": 33,
"grid_type": "",
"operatororganization": "",
"elevation": "",
"irradiationsensor_type": "",
"ownercontact": "rmb",
"weathersensor_md": "",
"tracker_type": "",
"oftakerorganization": "",
"dataloggers": [
 {
 "datalogger_num": 1,
 "datalogger_man": "Growatt",
 "datalogger_md": ""
 }
],
"financierorganization": "",
"peak_power": 20,
```

```

 "notes": "",
 "arrays": [
 {
 "num_modules": 0,
 "module_md": "",
 "module_man": "Growatt"
 }
]
},
"error_code": 0,
"error_msg": ""
}
` ``

```

**\*\*Return parameter description\*\***

| Parameter Name | Type   | Description                                                                |
|----------------|--------|----------------------------------------------------------------------------|
| name           | string | power station name                                                         |
| user_id        | int    | User ID to which the plant belongs                                         |
| country        | string | country name                                                               |
| longitude      | string | Longitude                                                                  |
| latitude       | string | Latitude                                                                   |
| create_date    | string | Building date                                                              |
| image_url      | string | Picture url                                                                |
| peak_power     | string | peak power (kWp)                                                           |
| currency       | string | currency unit                                                              |
| timezone       | string | Time Zone                                                                  |
| plant_type     | int    | Plant type(0-Residential Plant,1-Commercial Plant,2-Ground-Mounted Plants) |
| address1       | string | Plant address                                                              |



|error\_code|int|0: Normal return, 10001: System error, 10002: Power station does not exist, 10003: Power station ID is empty, 10004: User does not exist

|error\_msg|string| Error message prompt

**\*\*Remarks\*\***

- Get the frequency once every 5 minutes

**\*\*A brief description:\*\***

- Interface to get an overview of a plant data

**\*\*Request URL: \*\***

- `http(s)://test.growatt.com/v1/plant/data`

**\*\*Request method: \*\***

- get

**\*\*Parameters: \*\***

| Parameter Name | Whether Required | Type | Description |
|----------------|------------------|------|-------------|
|----------------|------------------|------|-------------|

|       |      |        |        |
|-------|------|--------|--------|
| :---- | :--- | :----- | :----- |
|-------|------|--------|--------|

|          |     |     |                  |
|----------|-----|-----|------------------|
| plant_id | Yes | int | Power Station ID |
|----------|-----|-----|------------------|

**\*\*Back to example\*\***

`,`

{

"data": {

"peak\_power\_actual": 20,

"monthly\_energy": "15.7",

"last\_update\_time": "2018-12-13 11:06:34",

```

 "current_power": 0,
 "timezone": "GMT+8",
 "yearly_energy": "15.7",
 "today_energy": "0",
 "carbon_offset": "9.4",
 "efficiency": "",
 "total_energy": "15.7"
 },
 "error_code": 0,
 "error_msg": ""
}
` ``

```

**\*\*Return parameter description\*\***

| Parameter Name    | Type   | Description                                                                                                  |
|-------------------|--------|--------------------------------------------------------------------------------------------------------------|
| peak_power_actual | string | Actual Peak Power (kW)                                                                                       |
| last_update_time  | string | Last received data                                                                                           |
| current_power     | string | current power (kW)                                                                                           |
| today_energy      | string | Power Generation (kWh)                                                                                       |
| monthly_energy    | string | Power Generation in the Month (kWh)                                                                          |
| yearly_energy     | string | Power generation in the current year (kWh)                                                                   |
| total_energy      | string | Cumulative power generation (kWh)                                                                            |
| carbon_offset     | int    | Equivalent reduction of CO2 emissions                                                                        |
| timezone          | string | Time Zone                                                                                                    |
| error_code        | string | 0: Normal return, 10001: System error, 10002: Power station does not exist, 10003: Power station ID is empty |
| error_msg         | string | Error Message Prompt                                                                                         |

**\*\*Remarks\*\***

- Get the frequency once every 5 minutes

**\*\*A brief description:\*\***

- Obtain an interface for historical power generation of a power station

**\*\*Request URL: \*\***

- `http(s)://test.growatt.com/v1/plant/energy`

**\*\*Request method: \*\***

- get

**\*\*Parameters: \*\***

| Parameter Name | Whether Required | Type | Description |
|----------------|------------------|------|-------------|
|----------------|------------------|------|-------------|

| :---- | :--- | :----- | :----- |
|-------|------|--------|--------|
|-------|------|--------|--------|

|          |     |     |                  |
|----------|-----|-----|------------------|
| plant_id | Yes | int | Power Station ID |
|----------|-----|-----|------------------|

|            |     |        |            |
|------------|-----|--------|------------|
| start_date | Yes | string | Start Date |
|------------|-----|--------|------------|

|          |     |        |                                              |
|----------|-----|--------|----------------------------------------------|
| end_date | Yes | string | End Date(date interval cannot exceed 7 days) |
|----------|-----|--------|----------------------------------------------|

|           |     |        |                                      |
|-----------|-----|--------|--------------------------------------|
| time_unit | Yes | string | time type:"day" or "month" or "year" |
|-----------|-----|--------|--------------------------------------|

|      |    |     |                       |
|------|----|-----|-----------------------|
| page | No | int | page number,default 1 |
|------|----|-----|-----------------------|

|         |    |     |                                               |
|---------|----|-----|-----------------------------------------------|
| perpage | No | int | Number of pages per page, default 20, max 100 |
|---------|----|-----|-----------------------------------------------|

**\*\*Back to example\*\***

...

{

  "data": {

    "count": 6,

    "time\_unit": "day",

    "energys": [

```
{
 "date": "2018-12-12",
 "energy": "0"
},
{
 "date": "2018-12-13",
 "energy": "7.6"
},
{
 "date": "2018-12-14",
 "energy": "0"
},
{
 "date": "2018-12-15",
 "energy": "0"
},
{
 "date": "2018-12-16",
 "energy": "0"
},
{
 "date": "2018-12-17",
 "energy": "0"
}
]
},
"error_code": 0,
"error_msg": ""
}
...
```

**\*\*Return parameter description\*\***

| Parameter Name | Type | Description |
|----------------|------|-------------|
|----------------|------|-------------|

|       |       |       |
|-------|-------|-------|
| ----- | ----- | ----- |
|-------|-------|-------|

|       |     |               |
|-------|-----|---------------|
| count | int | Total Records |
|-------|-----|---------------|

|           |        |                      |
|-----------|--------|----------------------|
| time_unit | string | time_unit on request |
|-----------|--------|----------------------|

|      |        |      |
|------|--------|------|
| date | string | date |
|------|--------|------|

|        |        |                        |
|--------|--------|------------------------|
| energy | string | Power Generation (kWh) |
|--------|--------|------------------------|

|            |     |                                                                                                                                               |
|------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|
| error_code | int | 0: Normal return, 10001: System error, 10002: Power station does not exist, 10003: Power station ID is empty, 10004: Time format is incorrect |
|------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|

|           |        |                      |
|-----------|--------|----------------------|
| error_msg | string | Error Message Prompt |
|-----------|--------|----------------------|

**\*\*Remarks\*\***

- Get the frequency once every 5 minutes
- This interface is only allowed to call 10 times a day

**\*\*A brief description:\*\***

- Obtain an interface for power data of a certain power station

**\*\*Request URL: \*\***

- ` http(s)://test.growatt.com/v1/plant/power `

**\*\*Request method: \*\***

- get

**\*\*Parameters: \*\***

| Parameter Name | Whether Required | Type | Description |
|----------------|------------------|------|-------------|
|----------------|------------------|------|-------------|

|       |      |        |       |
|-------|------|--------|-------|
| ----- | ---- | ------ | ----- |
|-------|------|--------|-------|

|plant\_id |yes |int |Power Station ID |  
|date |yes |string |date |

**\*\*Back to example\*\***

```
...
{
 "data": {
 "powers": [
 {
 "time": "2018-12-13 10:45",
 "power": 7903.39990234375
 },
 {
 "time": "2018-12-13 10:40",
 "power": 7689.7001953125
 }
],
 "count": 2
 },
 "error_code": 0,
 "error_msg": ""
}
...
```

**\*\*Return parameter description\*\***

|Parameter Name|Type|Description|  
|:----- |:-----|----- |  
|count |int | Total Records |

|time |string |Time |

|power |string |Power (W) |

|error\_code |int |0: Normal return, 10001: System error, 10002: Power station does not exist, 10003: Power station ID is empty or time format is incorrect |

|error\_msg |string | Error Message Prompt |

**\*\*Remarks\*\***

- Get the frequency once every 5 minutes

**\*\*A brief description:\*\***

- Obtain an interface for the plant information of a device

**\*\*Request URL: \*\***

- ` http(s)://test.growatt.com/v1/plant/sn\_plant `

**\*\*Request method: \*\***

- post

**\*\*Parameters: \*\***

|Parameter Name|Whether Required|Type|Description|

|:---- |:---|:----- |----- |

|device\_sn |Yes |string |Device serial number (non-collector serial number) |

**\*\*Back to example\*\***

```\n

{

```

"data": {
  "plant": {
    "status": 4,
    "locale": "",
    "total_energy": 15.699999809265137,
    "operator": "API interface test vendor",
    "country": "China",
    "city": "Shenzhen",
    "current_power": "",
    "create_date": "2018-12-12",
    "image_url": "2.png",
    "plant_id": 24765,
    "name": "API interface test power station",
    "longitude": "113.9",
    "user_id": 33,
    "installer": "API interface test vendor",
    "latitude": "22.6",
    "peak_power": 20,
    "latitude_d": null,
    "latitude_f": null
  }
},
"error_code": 0,
"error_msg": ""
}
, , ,

```

****Return parameter description****

| Parameter Name | Type | Description |
|----------------|------|-------------|
|----------------|------|-------------|

| | | |
|-------|-------|-------|
| ----- | ----- | ----- |
|-------|-------|-------|

|plant_id |int |Power Station ID |
|name |string |Power Station Name |
|user_id |int | User ID to which the plant belongs |
|country |string |country name |
|longitude |string |Longitude |
|latitude |string | Latitude |
|peak_power |string |peak power (kWp) |
|create_date |string |Building Date |
|installer |string |installer name |
|total_energy |string |Cumulative power generation (kWh) |
|count |int | Total power stations |
|latitude_d |string |Agent Name
|latitude_f |string |Agent Area
|error_code |int |0: Normal return, 10001: System error, 10002: Account does not exist,
10003: Device serial number is empty, 10004: Device type is empty, 10005: Device does
not exist |
|error_msg |string | Error Message Prompt |

****Remarks****

- Get the frequency once every 5 minutes